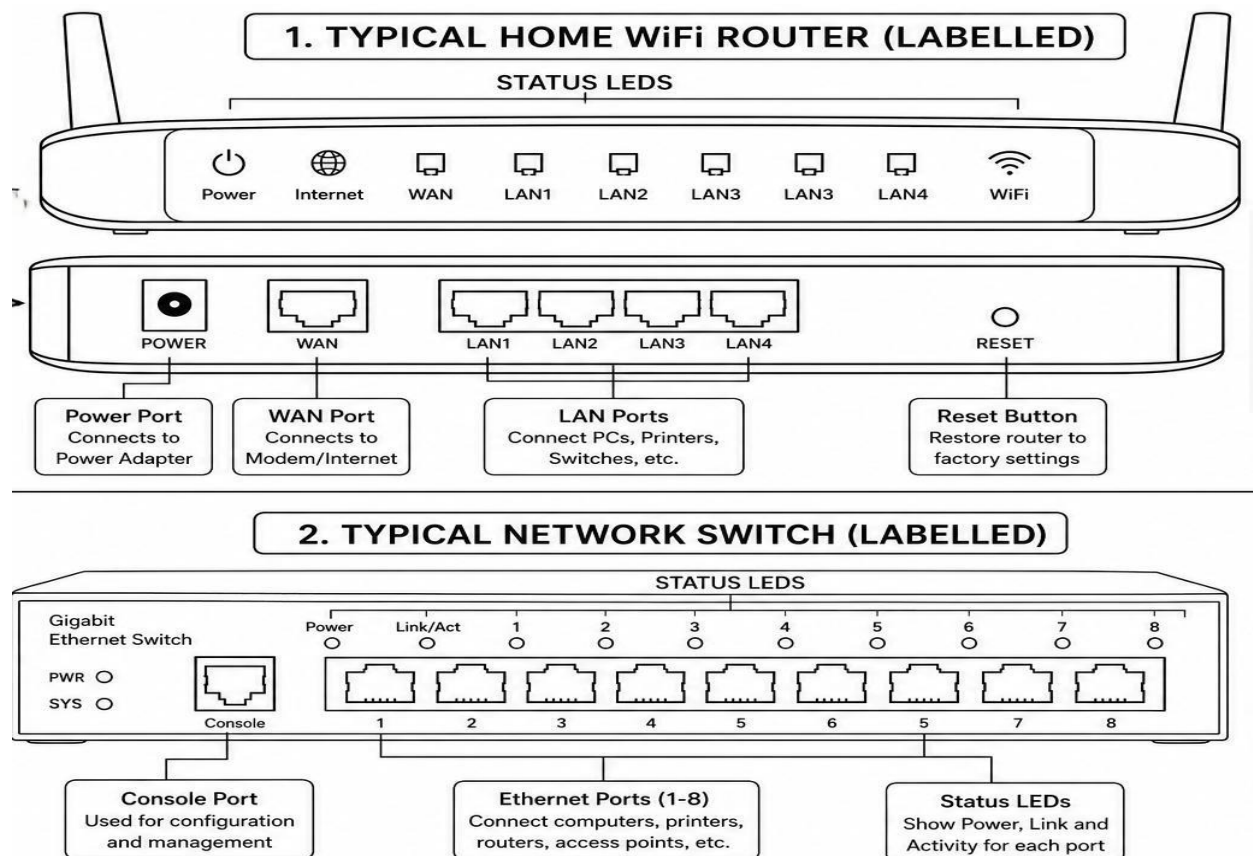


Day 2: Routers and Switches Theory :Basic functions and differences between routers and switches. Practical :Identify ports and LEDs on a router and switch.

Tasks :-

1.Draw a labelled diagram of a typical home WiFi router and a network switch, clearly marking at least 3 different ports (such as WAN, LAN, Console) and the location of status LEDs.

Ans. :-



2. List 3 key differences between routers and switches in terms of their main functions and how they handle network traffic.
Hint: Think about how each device deals with IP addresses, MAC addresses, and the types of networks they connect.

Ans. :-

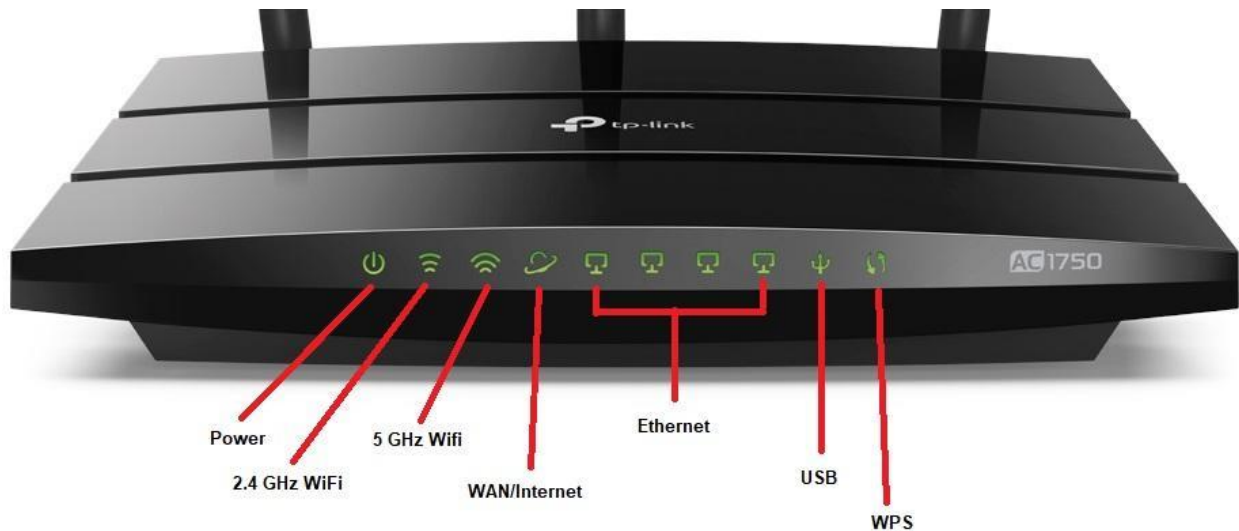
Differences :-

Aspect	Router	Switch
1. Main Function	Connects different networks (e.g., a home network to the Internet).	Connects multiple devices within the same Local Area Network (LAN).
2. Address Used	Uses IP addresses to determine the best path for data.	Uses MAC addresses to deliver data to the correct device.
3. Network Layer	Operates at the Network Layer (Layer 3) of the OSI model.	Operates at the Data Link Layer (Layer 2) of the OSI model.
4. Network Traffic	Routes traffic between different networks and selects the best route.	Forwards traffic only within the same network, reducing unnecessary data transmission.
5. Broadcast Domain	Separates broadcast domains, reducing network congestion.	By default, all ports are in the same broadcast domain (unless VLANs are configured).
6. Collision Domain	Each router interface is a separate collision domain.	Every switch port is its own collision domain, improving performance.
7. Internet Connection	Provides access to the Internet by connecting the LAN to a WAN/ISP.	Does not provide Internet access by itself; it requires a router.
8. Speed	Generally slower than a switch because it performs routing and security functions.	Usually faster for local data transfer because it only switches frames within the LAN.
9. Security Features	Can provide firewall, NAT, VPN, and traffic filtering features.	Offers limited security features, mainly port security and VLAN support (on managed switches).
10. Common Use	Used in homes, offices, and enterprises to connect different networks.	Used in homes, schools, and offices to connect computers, printers, and other devices.

3. Take a real or virtual image of a router and a switch (from your home, lab, or Google Images), and identify the following on each: Power port, at least 2 Ethernet ports, and any LED indicators. Annotate the image or describe their position and purpose.

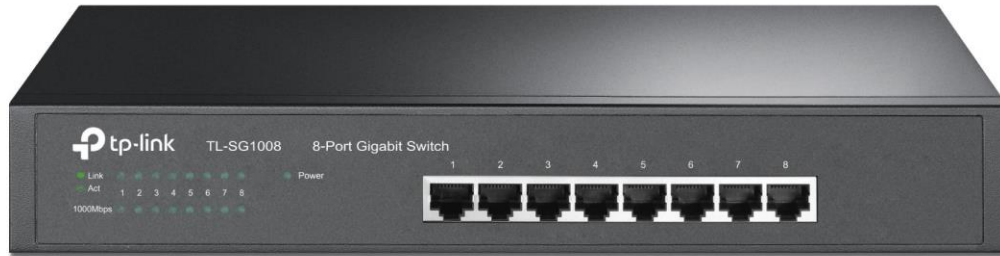
Ans. :-

Router :-



Component	Position	Purpose
Power Port	On the back panel, usually at the far left or right	Connects the power adapter to turn on the router.
Ethernet Port 1 (WAN/Internet)	Back panel, usually a blue or differently colored port	Connects the router to the modem or Internet connection.
Ethernet Port 2 (LAN)	Back panel, one of the yellow LAN ports	Connects a computer, printer, or another network device.
LED Indicators	Front or top panel	Show Power, Internet, Wi-Fi, LAN activity, and system status.

Network Switch :-



Component	Position	Purpose
Power Port	Back or side of the switch	Connects the power adapter to supply electricity.
Ethernet Port 1	Front panel (Port 1)	Connects a computer, router, or other network device.
Ethernet Port 2	Front panel (Port 2)	Connects another computer or network device.
LED Indicators	Above or beside each Ethernet port and the power LED	Indicate power status, link status, and network activity for each port.

4. Imagine you are setting up a small gaming café network for IPL streaming and multiplayer matches. Write a short scenario explaining which device (router or switch) you would connect first to the ISP line, and how you would connect 8 gaming PCs to the network. Justify your choices.

Ans.:-

We connect the router first to the ISP line, which then connects to a network switch to link the 8 gaming PCs.

Here is how this setup works in a real-world gaming café scenario:

Connect ISP to Router :-

The internet cable coming from your Internet Service Provider (ISP) plugs directly into the WAN (Wide Area Network) port of your router. The router acts as the gateway to the internet, managing incoming data traffic, providing firewall security, and assigning IP addresses to your devices via DHCP. A switch cannot do this; it cannot route public internet data to multiple private devices on its own.

Connect Router to Switch :-

Run a single Ethernet cable from one of the LAN (Local Area Network) ports on your router into any open port on an 8-port or 16-port Gigabit network switch. This extends the network and multiplies your available physical connection points.

Connect PCs to Switch :-

Plug each of the 8 gaming PCs into the remaining ports of the network switch using highspeed Cat6 Ethernet cables.

My setup will be like this :-

